

CO2 ASSESSMENT TOOL 2 - ARCHITECTURE DESIGN ASSIGNMENT

**AI-Driven Crop Disease Prediction and Precision
Farming Assistant**

Software Architecture Design Report

Course Code & Title:	CSA10 - Software Engineering
Course Outcome Assessed:	CO2 (Architecture Design Assignment)
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1. Comprehensive Problem Statement & Domain Context

1.1 Industry Context & Domain Background

Modern agriculture is undergoing a paradigm shift driven by digital technologies, including Artificial Intelligence (AI), the Internet of Things (IoT), remote sensing via agricultural drones, and high-resolution computer vision. As global populations rise and arable land becomes increasingly constrained, agricultural systems must maximize crop yields while simultaneously minimizing environmental footprints. Precision farming has emerged as a crucial approach to achieve sustainable productivity by leveraging data-driven decision support tools for early disease detection, dynamic irrigation, and optimized resource allocation.

1.2 The Core Problem & Challenges in Traditional Agriculture

In traditional agricultural workflows, crop diseases are primarily identified through manual field inspection and physical expert consultation. This approach suffers from critical structural shortcomings:

- **Delayed Disease Detection:** Pathogens spread unseen during early infection stages. By the time visible necrotic spots, leaf discolouration, or wilting appear, the disease has often reached an epidemic threshold, causing severe yield losses.
- **Overuse of Agrochemicals:** Fearing catastrophic crop loss, farmers frequently apply broad-spectrum chemical pesticides reactively and indiscriminately. This drastically increases input costs, damages soil micro-ecology, pollutes local aquatic ecosystems, and leads to chemical residue build-up.
- **Sub-optimal Resource Utilization:** Traditional irrigation and fertilization rely on static calendar schedules or rule-of-thumb estimates rather than real-time soil moisture, pH, N-P-K nutrient availability, and ambient evapotranspiration rates. This leads to either crop water stress or wasteful over-irrigation.
- **Expert Accessibility Gap:** Smallholder and rural farmers face significant geographic and economic barriers in accessing agricultural extension experts, leading to misdiagnoses and improper chemical selection.

1.3 Proposed AI-Driven Solution

To address these challenges, the proposed AI-Driven Precision Farming Assistant acts as an end-to-end intelligent agricultural decision platform. The platform ingests multi-modal data streams—including smartphone and drone crop imagery, continuous IoT soil telemetry (moisture, temperature, pH, N-P-K levels), and hyper-local meteorological forecasts. By synthesizing computer vision models with real-time agronomic rules engines, the platform delivers early disease diagnoses, prescriptive treatment advisories, optimized dynamic irrigation schedules, and comprehensive farm analytics to drive sustainable agriculture.

2. System Requirements Analysis

2.1 High-Level System Objectives

- **Early Pathogen Detection:** Achieve automated early-stage plant pathology identification using deep learning vision models.
- **Prescriptive Agrochemical Advisory:** Provide actionable, eco-friendly crop treatment measures with precise chemical and bio-pesticide dosage instructions.
- **Precision Resource Optimization:** Dynamically adjust irrigation and fertigation schedules based on live IoT sensor streams and predictive weather data.
- **Longitudinal Analytics:** Deliver actionable historical crop health insights and regional analytics to support sustainable farming practices.

2.2 Functional Requirements (FRs)

- **FR-1: Image Ingestion & Disease Diagnostics** — The system shall allow farmers and drone operators to upload multi-angle RGB imagery of crop leaves/stems and process them through Convolutional Neural Networks (CNNs) to diagnose diseases with confidence ratings and bounding-box localized symptoms.
- **FR-2: Multi-Modal Sensor Ingestion** — The system shall ingest real-time MQTT telemetry from field IoT nodes (soil moisture, temperature, pH, N-P-K levels) and external weather API feeds at configurable intervals.
- **FR-3: Prescriptive Treatment Engine** — Upon identifying a disease or nutrient deficiency, the system shall generate a structured intervention plan detailing organic/chemical treatment options, exact dosage calculations, and application safety protocols.
- **FR-4: Dynamic Irrigation Scheduling** — The system shall compute real-time Reference Evapotranspiration (ET0) to generate dynamic, automated irrigation schedules pushed directly to smart valve actuators or farmer mobile apps.
- **FR-5: Analytics & Advisory Dashboard** — The platform shall provide interactive dashboards displaying spatial disease distribution maps, historical health indices, and yield forecast reports.

3. Quality Attributes & Non-Functional Requirements (NFRs)

Architectural design decisions are guided by strict quality attribute requirements tailored to the agricultural domain:

3.1 Key Architectural Quality Attributes

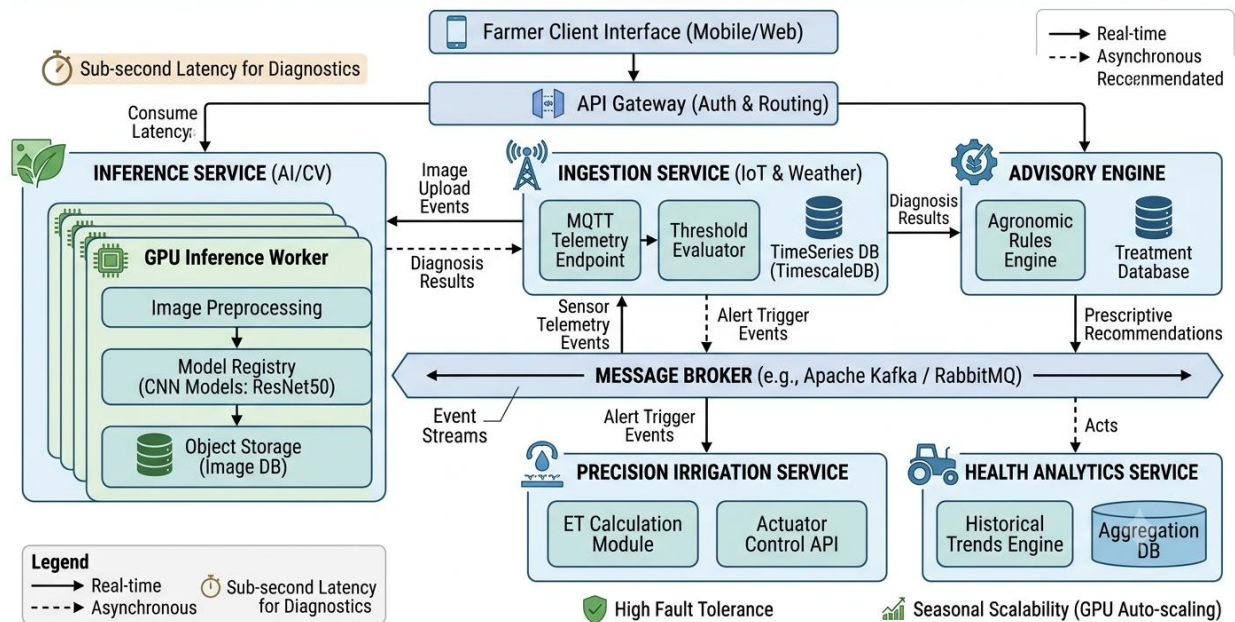
- **Scalability (Horizontal Elasticity):** The system must seamlessly handle seasonal spikes (e.g., planting/harvesting seasons) across thousands of registered farms and millions of daily IoT data points without performance degradation.
- **Reliability & Fault Tolerance:** The mobile application must support offline-first data capture and local model inference (via TFLite), caching sensor logs and images during rural connectivity outages and synchronizing automatically upon reconnection.
- **Availability (24/7 Operational Uptime):** The cloud backend must achieve 99.9% operational availability during critical agricultural cycles, ensuring uninterrupted access to real-time irrigation triggers and disease alerts.
- **Performance & Latency:** Image disease classification latency must not exceed 2.0 seconds for cloud processing and 500 milliseconds for edge mobile processing. Telemetry stream processing must maintain sub-second response times.
- **Security & Data Privacy:** All data in transit must be encrypted via TLS 1.3, and data at rest encrypted using AES-256. Strict Role-Based Access Control (RBAC) must protect farmer identity and field telemetry data.
- **Maintainability & Modularity:** The platform must adopt modular microservices allowing independent neural network model retraining, database updates, and service deployments without requiring system downtime.

4. Software Architecture Selection & Technical Justification

4.1 Selected Architectural Style: Event-Driven Microservices Architecture

To meet the multi-modal ingestion, high scalability, and heavy AI compute demands of the platform, an Event-Driven Microservices Architecture (with Hybrid Edge-Cloud Processing) was selected.

EVENT-DRIVEN MICROSERVICES ARCHITECTURE: AI-DRIVEN PRECISION FARMING ASSISTANT

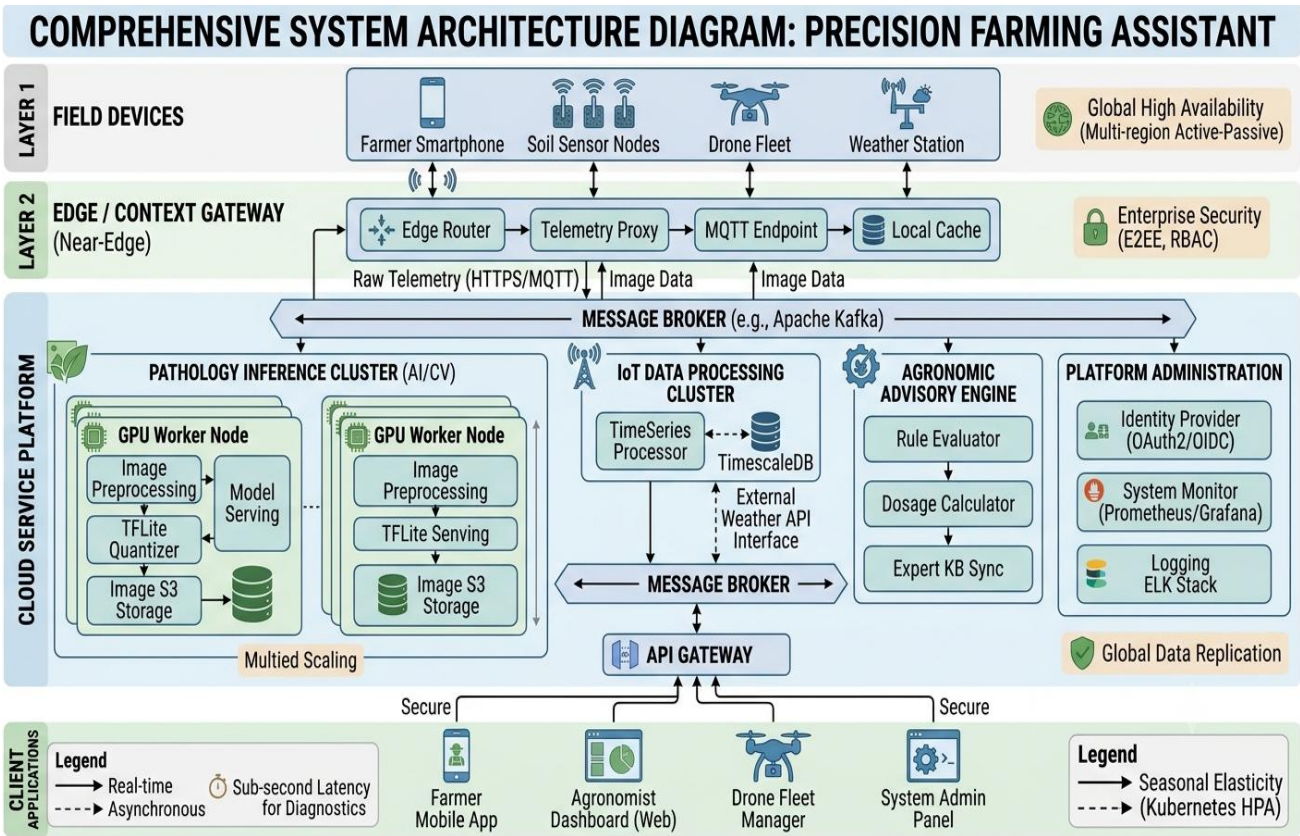


4.2 Architectural Justification

- Asynchronous Event Handling:** IoT soil sensors stream data continuously. An event bus (Apache Kafka) decouples high-frequency telemetry producers from consumer microservices, eliminating processing bottlenecks.
- Isolated Compute Workloads:** Computer vision model inference requires GPU-intensive nodes, while telemetry ingestion requires lightweight, high-I/O compute nodes. Microservices allow independent auto-scaling of specific infrastructure workloads.
- Resilience & Fault Isolation:** If the historical analytics service experiences a downtime, core operational paths—such as real-time disease detection and irrigation alerts—continue operating unaffected.

5. Comprehensive System Architecture Diagram

The system architecture illustrates the complete end-to-end topology, from field-level IoT sensors and smartphone clients up to cloud-native microservices and multi-model database storage:



6. Detailed Architectural Component Analysis

6.1 Core System Services

- **API Gateway & Edge Router:** Acts as the single secure entry point for all client devices. Enforces TLS 1.3 security, manages OAuth2/JWT authentication, executes rate limiting, and routes traffic to downstream services.
- **Crop Pathology & Disease Analysis Service:** Receives crop imagery, executes image normalization/preprocessing, and runs inference on trained Convolutional Neural Networks (ResNet50 / EfficientNet) to identify disease pathogens and severity levels.
- **IoT & Environmental Ingestion Service:** Processes persistent MQTT telemetry streams from soil sensor gateways and integrates hyper-local weather forecast updates via external REST APIs.
- **Prescriptive Advisory Engine:** Cross-references disease diagnostic outputs and soil telemetry against an expert agronomic rules database to generate organic and chemical intervention recommendations.
- **Precision Irrigation Scheduler:** Evaluates soil moisture levels against Penman-Monteith Evapotranspiration (ET_0) formulas to automatically calculate precise crop water requirements and schedule irrigation runs.

7. Component Interaction Patterns & Workflows

7.1 Workflow 1: Crop Disease Diagnostic Sequence

```
[ Farmer App ] ---> (1) Upload Image (HTTPS POST) -----> [ API Gateway ]
[ API Gateway ] -> (2) Validate Token & Forward -----> [ Disease Service ]
[ Disease Svc ] -> (3) Execute Model Inference (GPU) -----> [ AI Inference Engine ]
[ Inference ]   -> (4) Return Disease ID & Confidence Score -> [ Disease Service ]
[ Disease Svc ] -> (5) Query Treatment Measures -----> [ Advisory Engine ]
[ Advisory ]    -> (6) Return Dosage & Treatment Plan -----> [ Farmer App ]
```

7.2 Workflow 2: Automated IoT Sensor Telemetry & Irrigation Workflow

- **Step 1 (Sensor Transmission):** Field soil sensors publish telemetry payloads (moisture, pH, N-P-K) via MQTT to the API Gateway every 15 minutes.
- **Step 2 (Event Ingestion):** The API Gateway validates payload signatures and pushes telemetry into the Kafka topic `sensor-telemetry-stream`.
- **Step 3 (Processing & Storage):** The IoT Microservice consumes the message, writes records to TimescaleDB, and evaluates moisture thresholds.
- **Step 4 (Automated Advisory Trigger):** If soil moisture falls below the crop critical wilting point, an event trigger alerts the Irrigation Scheduler to dispatch a dynamic watering schedule.

8. Evaluation of Quality Attributes

The proposed architecture directly fulfills non-functional quality requirements as detailed below:

Quality Attribute	Architectural Strategy & Realization
Scalability	Microservices are deployed in auto-scaling Kubernetes clusters. GPU worker pools scale dynamically based on image diagnostic queue length.
Reliability	Kafka buffers telemetry during database outages. Mobile clients cache data locally in SQLite/Room and sync automatically upon network reconnection.
Performance	Sub-second image diagnostics achieved via TensorFlow Lite quantization on edge nodes, and Redis caching for frequent agronomic queries.
Security	End-to-end TLS 1.3 wire encryption, AES-256 disk encryption, OAuth2 access controls, and RBAC isolation for farm records.

9. Architectural Rationale & Trade-Off Analysis

9.1 Trade-Off Analysis

- **Microservices vs. Monolith:** Microservices introduce operational overhead (Kubernetes management, distributed tracing, complex deployment pipelines). However, this trade-off is essential to allow GPU-heavy AI processing to scale independently without over-provisioning standard web servers.
- **Eventual Consistency vs. Immediate Consistency:** Telemetry processing uses eventual consistency through Kafka streams to maximize ingestion speed, while critical user account and advisory management use strong consistency in relational databases.
- **Cloud AI vs. Edge AI Execution:** Running full deep neural networks in the cloud guarantees maximum diagnostic accuracy, while lightweight TensorFlow Lite models on mobile devices provide rapid offline diagnostics at slightly reduced accuracy.

10. Future Scope & Sustainable Agriculture Support

10.1 Future Technical Enhancements

- **Satellite Imagery Integration:** Integrate Sentinel-2 and Landsat satellite multi-spectral imagery to monitor field-wide Normalized Difference Vegetation Index (NDVI) health scores.
- **Autonomous Drone Fleet Coordination:** Deploy Edge-TPU micro-controllers on autonomous spray drones to perform real-time localized spot-spraying based on live vision inference.

10.2 Supporting Sustainable Agricultural Practices

- **Reduced Chemical Contamination:** Targeted, early treatment advisories prevent blanket pesticide application, significantly reducing chemical runoff into soil and local waterways.
- **Freshwater Conservation:** Dynamic evapotranspiration scheduling ensures crops receive exact required water quantities, preventing over-irrigation and protecting freshwater reserves.

11. Summary Conclusion & References

11.1 Conclusion

The proposed Event-Driven Microservices Architecture provides a robust, scalable, and highly available foundation for the AI-Driven Crop Disease Prediction and Precision Farming Assistant. By decoupling high-frequency IoT sensor telemetry ingestion from GPU-intensive computer vision pipelines, the architecture guarantees real-time responsiveness, offline resilience, and long-term maintainability. This platform effectively empowers farmers with actionable early disease diagnostics and precision irrigation scheduling, promoting agricultural productivity and environmental sustainability.

11.2 References

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